Discussion

Tuesday, October 13, 2020 6:06 PM

Logistics: T 10-11am
Fr 8-9 am

OH → CCLE.

Example 1.1. **(PSI 1.1.1)** Of a group of patients having injuries, 28% visit both a physical therapist and a chiropractor and 8% visit neither. Say that the probability of visiting a physical therapist exceeds the probability of visiting a chiropractor by 16%. What is the probability of a randomly selected person from this group visiting a physical therapist?

A) 0.24 B) 0.38 C) 0.52 $\boxed{\text{D) } 0.68}$ E) 0.72

(Sol 1) let P be the randomly chosen patien.

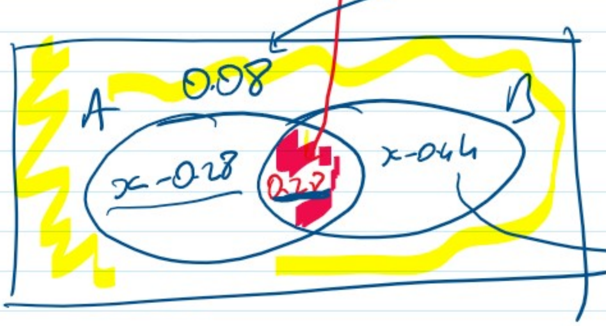
$A = \{ P \text{ visits PT} \}$

$B = \{ P \text{ visits Ch} \}$

- $P(A \cap B) = 0.28$
  ↳ P visits both.
- $P((A \cup B)') = 0.08$
- $P(A) = P(B) + 0.16$

$P(A) = ?$

$P(A) = x$



$P(B) = P(A) - 0.16$
$= x - 0.16$

→ $P(B) - 0.28 = x - 0.16 - 0.28$
$= x - 0.44$

$0.08 + (x - \cancel{0.28}) + (\cancel{0.28}) + (x - 0.44) = 1$

$2x - 0.36 = 1$

$$2x = 1.36$$
$$x = 0.68$$

Sol 2: • $P(A \cap B) = \underline{0.28}$ • $P((A \cup B)') = 0.08$

↳ P visits both.

• $P(A) = P(B) + 0.16$ • $P(A) = ?$

$P((A \cup B)') = 0.08$

↳ $1 - P(A \cup B) = 0.08 \Rightarrow P(A \cup B) = 0.92$

PIE : $P(A \cup B) = P(A) + \underbrace{P(B)}_{P(A) - 0.16} - \underbrace{P(A \cap B)}_{0.28}$

$$0.92 = 2P(A) - 0.44$$
$$0.92 + 0.44 = 2P(A)$$
$$1.36 = 2P(A) \Rightarrow P(A) = 0.68$$

Ch 1.2

• If we have n different object, there are in total

$$n! = n.(n-1) \cdots 1$$

different ways to permute them.

• If we want to permute r of the n different objects, then there are at total

$n!$

there are at total

$$n\cdot(n-1)\cdots(n-r+1)=\frac{n!}{(n-r)!}={}_nP_r$$

EX: 10 horses are racing in a derby. The first, the second and the third will be determined on recorded. How many different ways are possible?

A) 120 B) 360 C) 540 D) 720 E) 10!

Sol: We have 10 different horses, we want to permute 3 of them.

$$\underset{1^{st}}{\underline{10}}\ \underset{2^{nd}}{\underline{9}}\ \underset{3^{rd}}{\underline{8}}=\frac{10!}{7!}=720$$

$$={}_{10}P_3$$

• (Combination) Assume we have n different objects, we want to choose r of them (ordering does not matter). Then there are exactly

$${}_nC_r=\binom{n}{r}=\frac{n!}{r!\,(n-r)!}$$

different way to do it.

EX: We have 10 different flavored candies, and we want to choose 3 of them and put them in a bag. How many ways this is possible?

A) 120 B) 240 C) 360 D) 720 E) 10!

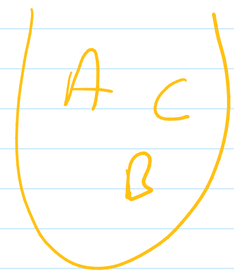
Sol: $${}_{10}C_3=\frac{10!}{3!\cdot 7!}=\frac{10\cdot 9\cdot 8\cdot \cancel{7!}}{3!\cdot \cancel{7!}}=\frac{10\cdot 9\cdot 8}{6}=120$$

Sol: $_{10}C_3 = \dfrac{10!}{3!\cdot 7!} = \dfrac{10\cdot 9\cdot 8\cdot \cancel{7!}}{3!\cdot \cancel{7!}} = \dfrac{10\cdot 9\cdot 8}{6} = 120$

$\underline{10}\quad \underline{9}\quad \underline{8} \rightarrow 720.$

A	B	C
A	C	B
B	A	C
B	C	A
C	A	B
C	B	A

3! ways to permute.



$\dfrac{720}{3!}$ is the answer.

Ex: (1-2-5). How many four letter code words are possible using the letters in IOWA if

(a) the letters may not be repeated? A) 1 B) 6 C) 24 D) 64 E) 256

(b) " " " be repeated.? A) 1 B) 6 C) 24 D) 64 E) 256

Sol: 4 different letters

$\underline{4}\quad \underline{3}\quad \underline{2}\quad \underline{1} = 4! = 24$

(b)

$\underline{4}\quad \underline{4}\quad \underline{4}\quad \underline{4} = 4^4 = 256$

(Ex) Code words can be up to 5 letters. How many code words are possible if "IOWA"

(a) letters may not be repeated. A) 24 B) 48 C) 64 D) 256 E) 1024

(b) " " be " A) 24 B) 120 C) 1024 D) 1364 E) 2048

AW WIO A AAO

Sol: (a)

$\underline{4} \rightarrow 4 \quad ({}_4P_1)$

$\underline{4}\ \underline{3} \rightarrow 12 \quad ({}_4P_2)$

$\underline{4}\ \underline{3}\ \underline{2} \rightarrow 24 \quad ({}_4P_3)$

$\underline{4}\ \underline{3}\ \underline{2}\ \underline{2} \rightarrow 24 \quad ({}_4P_4 = 4!)$

$\underline{4}\ \underline{3}\ \underline{2}\ \underline{2}\ \underline{0} \rightarrow 0$

$4 + 12 + 24 + 24 = 64$

(b)

$\underline{4} \rightarrow 4$

$\underline{4}\ \underline{4} \rightarrow 4^2 = 16$

$\underline{4}\ \underline{4}\ \underline{4} \rightarrow 4^3 = 64$

$\underline{4}\ \underline{4}\ \underline{4}\ \underline{4} = 4^4 = 256$

$\underline{4}\ \underline{4}\ \underline{4}\ \underline{4}\ \underline{4} \rightarrow 4^5 = 1024$

Adding them up give the answer $\boxed{1364.}$